Evaluating RAG-Based Course Advising Chatbots Using LLM-as-a-Judge Methodology

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Abstract

Retrieval-Augmented Generation (RAG) chatbots are increasingly deployed for academic advising, yet it remains unclear whether these systems can provide reliable, hallucination-free guidance. We investigate this question by evaluating a production course advising chatbot using LLM-as-a-judge methodology with Gemini 3 Flash. To enable systematic assessment, we develop AdvisingBench, a benchmark comprising 55 expert-evaluated questions across three complexity categories (common, specific and edge cases). We evaluate the system on four metrics: answer relevancy, faithfulness, context relevancy and answer correctness. Our results reveal performance variation across query categories, with edge cases showing 6% higher overall scores than common queries despite lower context relevancy. We find that context relevancy is the metric most sensitive to query complexity, while answer relevancy remains consistently high (>0.95) across all categories. Notably, hallucination failures were rare (3 of 53 cases), suggesting that RAG architectures substantially constrain models to retrieved content in academic advising contexts, though they do not eliminate fabrication risk. This work provides initial empirical evidence on whether AI can support course advising, and offers a methodology and benchmark for future validation studies.

CCS Concepts

• **Information systems** → **Question answering**; • **Computing methodologies** → *Natural language generation*; • **Applied computing** → *Education*.

Keywords

RAG evaluation, LLM-as-a-judge, academic advising, course recommendation, educational AI, benchmark

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1 Introduction

Large Language Models (LLMs) integrated with Retrieval-Augmented Generation (RAG) architectures are transforming academic support services in higher education [8]. These systems promise 24/7 availability, consistent responses, and reduced burden on human advisors. However, deploying AI for academic advising carries significant responsibility as incorrect course recommendations can delay graduation, increase costs and cause students stress. A critical question therefore arises: can RAG-based chatbots provide reliable, hallucination-free academic guidance? Despite growing adoption, standardized evaluation frameworks for academic advising chatbots remain notably absent. A recent comprehensive survey of RAG chatbots in education reported that there is currently no widely accepted standard for evaluating such systems and highlighted this as a major weakness in the field [11]. Existing systems report heterogeneous metrics that prevent cross-system comparison: AdvisingWise achieved high-accuracy responses on 84.2% (16/19) of evaluable advising queries in an expert-validated sample of 20 queries [7], NileAdvisor reported 0.93 query accuracy using knowledge graphs [1], while CourseQ focused more on usability and user-perceived accuracy than on formally measuring factual correctness of its recommendations [10]. This heterogeneity leaves practitioners without guidance for selecting among competing approaches or understanding how system performance varies across different types of student queries. The RAGAS framework [3] has emerged as a de facto standard for RAG evaluation, offering metrics for faithfulness, answer relevancy and context quality. However, practical validation reveals concerning limitations: the TREC 2024 RAG Track evaluation found GPT-4o and human judgments match perfectly only 56% of the time in manual assessments, rising to 72% with post-editing [13]. These gaps suggest that automated evaluation, while useful for scale, requires careful validation. This paper addresses these gaps by evaluating a production RAG-based course advising chatbot to provide initial empirical characterization of system behavior across query types. We develop AdvisingBench, a 55-question benchmark with expert-evaluated reference answers across three complexity categories, and apply LLM-as-a-judge methodology to assess the system across four established metrics. We contribute: (1) initial empirical characterization of how a RAG-based academic advising system performs across query complexity categories; (2) analysis of how evaluation metrics behave differently across these categories; and (3) a reusable benchmark and methodology that can be validated against human advisor assessments in future work.

2 Related Work

To contextualize our evaluation approach, we review three relevant areas: RAG evaluation frameworks, educational AI chatbots, and LLM-as-a-judge methodology.

2.1 RAG Evaluation Frameworks

The RAGAS framework introduced reference-free metrics for RAG systems, including faithfulness (factual consistency with retrieved context), answer relevancy (appropriateness to query), and context precision/recall [3]. While widely adopted, practitioners report variable reliability. The TREC 2024 RAG Track evaluation found GPT-4o and human judgments match only 56% perfectly in manual assessments, rising to 72% with post-editing [13]. This suggests automated evaluation, while useful for scale, requires careful validation. Recent work on the CRAG benchmark reveals that "even state-of-the-art RAG solutions struggle with accuracy, especially on questions with higher dynamism, lower popularity, or higher complexity" [14].

2.2 Educational AI Chatbots

RAG chatbots for education have proliferated across domains. A systematic review identified common applications including tutoring, FAQ systems, and administrative support [11]. However, the review noted that many evaluation methods "lack comprehensiveness or detailed specification." For course advising specifically, published systems use inconsistent evaluation approaches. The AdvisingWise system was evaluated using a mixed-methods approach (expert evaluation, LLM-as-a-judge, and a user study with advisors) and adopted a conservative strategy of declining to answer when uncertain, which the authors argue helped avoid incorrect responses and contributed to high observed accuracy [7]. The From Interests to Insights system applied RAG to course descriptions using embedding similarity, but focused on recommendation relevance rather than factual accuracy [2]. An academic services chatbot at FPMIPA UPI achieved 0.91 faithfulness on factual queries but context recall dropped to 0.59 for reasoning tasks which shows a 35% performance gap [6]. These varied approaches underscore the need for systematic evaluation of whether AI can reliably support academic advising.

2.3 LLM-as-a-Judge Methodology

The LLM-as-a-judge paradigm uses capable language models to evaluate other model outputs [15]. The foundational MT-Bench study established that GPT-4 achieves agreement rates comparable to human-to-human agreement [15]. G-Eval extended this with chain-of-thought prompting, achieving 0.514 Spearman correlation on summarization tasks [9]. However, documented biases affect reliability. Position bias with GPT-4 exhibits approximately 35% inconsistency, while Claude-v1 shows approximately 76% as models may flip decisions when answer order is swapped [15]. Verbosity bias is more severe: Claude and GPT-3.5 preferred longer responses over 90% of the time when verbose distractors were added [15]. The authors observe that GPT-4, as a judge, gives GPT-4 about a 10% higher win rate than humans do, but they emphasize that the evidence for true self-preference bias is inconclusive. A recent ACM IUI study specifically found "limitations of the LLM-as-a-Judge approach for evaluating LLM outputs in expert knowledge tasks" [12]. Recent comprehensive benchmarking of 54 LLM judges found that model selection significantly impacts evaluation quality [5]. The Judge's Verdict study demonstrated that Gemini Flash variants achieve Tier 1 human-like performance, with gemini-2.5-flash-lite achieving a z-score of 0.17 on their human-likeness test, indicating judgment patterns nearly indistinguishable from human evaluators. Best practices for LLM-as-a-judge include detailed evaluation rubrics with explicit criterion anchors tied to discrete scale levels, decomposing evaluation into separable per-metric judgments, and using structured chain-of-thought reasoning to guide scoring [4, 9]. These approaches can improve consistency and help mitigate known judge biases such as length preference while improving alignment with human judgments [4, 9]. We incorporate these practices in our evaluation methodology.

3 Methodology

Building on the evaluation approaches reviewed above, we now describe our methodology for assessing whether a RAG-based course advising chatbot can provide reliable guidance.

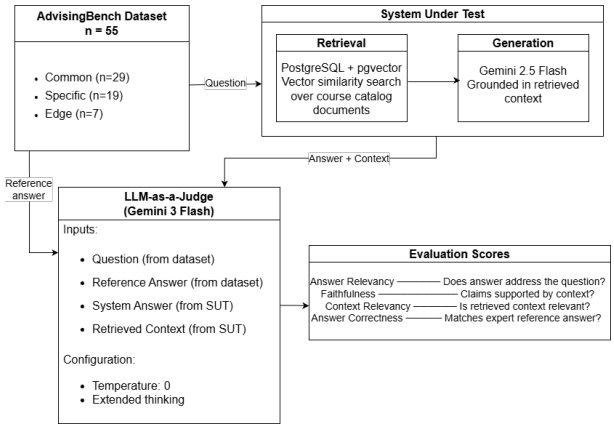


Figure 1: AdvisingBench evaluation pipeline: benchmark questions flow through the RAG system and are scored via LLM-as-a-judge methodology.

3.1 System Under Test

We evaluate a production RAG-based course advising chatbot deployed at a university in North America. The system assists students in Computer Science, Engineering, Mathematics, and Data Science programs with course planning queries. The architecture integrates:

- **Retrieval**: Vector similarity search over course catalog documents using PostgreSQL with pgvector extension
- **Generation**: Gemini 2.5 Flash for response synthesis
- **Knowledge Base**: Official university course descriptions, prerequisites, restrictions, and program requirements

The chatbot is instructed to answer strictly from retrieved context and politely decline out-of-scope queries.

3.2 Benchmark Dataset

To systematically assess the chatbot across different query types, we developed AdvisingBench, a dataset of 55 questions organized into three complexity categories based on expected retrieval and reasoning difficulty: **Common queries (n=29, 53%):** Frequent questions with straightforward answers directly stated in course descriptions. Examples include course descriptions, credit values, prerequisites, term offerings, and program information. These test baseline system performance on typical student queries. Example: "What are the prerequisites for CS 320?" **Specific queries (n=19, 35%):** Questions requiring interpretation of course policies or evaluation of eligibility. Examples include prerequisite chain analysis, credit exclusion scenarios, grade threshold evaluation, and co-requisite interpretation. These test the system's ability to reason about policy implications. Example: "If I scored 65% in CS 111, can I take CS 121? What if I scored 55%?" **Edge cases (n=7, 13%):** Adversarial, ambiguous, or out-of-scope queries testing system boundaries. Examples include requests for system prompt exfiltration, questions outside course advising scope, and ambiguous course references. These test appropriate refusal and escalation behavior. Example: "What are the prerequisites for 304?" Each question includes an expert-evaluated reference answer validated against official university academic calendar. Table 1 summarizes the category distribution and subcategories.

Table 1: AdvisingBench Dataset Composition

Category	Example Subcategories	Count
Common	course_description, prerequisites, term_offering, major_info	29
Specific	prereq_chain, credit_exclusion, grade_threshold, eligibility	19
Edge	out_of_scope, ambiguous, prompt_exfiltration	7
Total		**55**

3.3 Evaluation Metrics

We evaluate four metrics using an LLM-as-a-judge approach: **Answer Relevancy:** Does the answer appropriately address the question (or appropriately refuse/redirect if out-of-scope)? Scored 0.0-1.0. **Faithfulness:** Are all factual claims in the answer supported by the retrieved context? Penalizes hallucinations. Scored 0.0-1.0. **Context Relevancy:** Is the retrieved context relevant to the question? Evaluates retrieval quality independent of generation. Scored 0.0-1.0. **Answer Correctness:** How factually correct is the answer compared to the expert-evaluated reference answer? Score 1.0 if all key facts match, 0.5 if partially correct, 0.0 if incorrect. This reference-based metric follows standard QA evaluation practice [9]. **Overall:** Aggregate score computed by the judge model. The judge prompt suggests using a weighted average prioritizing faithfulness, answer correctness, and answer relevancy, but the specific weights are determined dynamically by the judge rather than fixed a priority. This reflects the relative importance of avoiding hallucinations and providing accurate information in academic advising contexts.

3.4 Judge Configuration

Following best practices for LLM-as-a-judge [9, 15], we configure the judge model with:

- **Model:** Gemini 3 Flash Preview with high thinking level
- **Temperature:** 0 for deterministic evaluation
- **Prompt Structure:** Detailed rubric with scoring criteria, followed by question, reference answer, system answer, and retrieved context
- **Output Format:** Structured JSON with scores and brief explanatory notes

We selected Gemini 3 Flash as the judge model based on recent benchmarking showing Gemini Flash variants achieve Tier 1 human-like judgment, with gemini-2.5-flash-lite exhibiting agreement patterns nearly indistinguishable from human evaluators [5]. While the benchmarking validated earlier Gemini Flash versions rather than 3 Flash Preview specifically, we extrapolate that architectural continuity within the Gemini Flash family preserves these evaluation characteristics. Temperature 0 ensures reproducible evaluation by eliminating sampling randomness, following standard practice for automated assessment [9]. We enabled the model's extended reasoning mode (high thinking level) to allow deliberation before scoring, as chain-of-thought approaches have been shown to improve correlation with human judgments [9]. The judge is instructed that out-of-scope queries should be appropriately refused, and that answers must be grounded in retrieved context to score well on faithfulness.

3.5 Experimental Procedure

For each test case, we: (1) submit the question to the course advising API; (2) capture the system response and retrieved context; (3) pass question, response, context, and reference answer to the judge; and (4) record all scores, latencies, and judge notes. The evaluation completed successfully on 53 of 55 test cases (96.4% completion rate).

4 Results

We now present the evaluation results, organized to address our research questions about metric variation across query categories and patterns that inform system reliability. We begin with aggregate performance, then examine category-level differences, and conclude with failure mode analysis.

4.1 Overall Performance

Table 2 presents aggregate performance across all successful test cases. The system achieves strong answer relevancy (0.979) and respectable answer correctness (0.758), with faithfulness (0.764) and context relevancy (0.702) showing more variation. These aggregate scores suggest reasonable overall performance, but examining results by query category reveals important variation.

4.2 Performance by Category

Table 3 reveals notable variation across complexity categories. Counter-intuitively, edge cases achieve the highest faithfulness (0.843) despite lowest context relevancy (0.643). This suggests the system appropriately grounds refusal responses in available context

Table 2: Overall Benchmark Performance (n=53)

Metric	Mean Score
Answer Relevancy	0.979
Faithfulness	0.764
Context Relevancy	0.702
Answer Correctness	0.758
Overall	**0.751**

Table 3: Performance by Query Category. Two test cases failed due to transient API errors: one common query and one specific query. These infrastructure failures were excluded from analysis.

Metric	Common (n=28)	Specific (n=18)	Edge (n=7)
Answer Relevancy	0.979	0.989	0.957
Faithfulness	0.754	0.750	0.843
Context Relevancy	0.693	0.739	0.643
Answer Correctness	0.739	0.778	0.786
Overall	**0.727**	**0.781**	**0.771**

while recognizing queries as out-of-scope. Specific queries outperform common queries on overall score (0.781 vs 0.727), driven by higher context relevancy (0.739 vs 0.693) and answer correctness (0.778 vs 0.739). This may reflect that specific queries require more targeted retrieval, resulting in more focused context.

4.3 Metric Correlation Analysis

To understand metric relationships, we examined score distributions across categories. Answer relevancy shows minimal variation (range: 0.032), remaining consistently high regardless of query complexity. This suggests it may be less discriminative for identifying system weaknesses. Context relevancy shows the widest variation by category (range: 0.096), with specific queries retrieving more relevant context than both common and edge cases. This metric appears most sensitive to query-retrieval alignment. Faithfulness shows moderate variation (range: 0.093), with highest scores on edge cases where the system correctly declines to answer based on irrelevant retrieved context.

4.4 Failure Mode Analysis

Examining cases with overall scores below 0.6, we identified four primary failure patterns: **Retrieval misalignment (n=6):** Questions retrieved tangentially related courses rather than the specific course queried. Example: asking about CS 328 prerequisites retrieved generic CS course listings without the specific course. Other instances included queries about CS 441, DATA 311, MATH 221, and MATH 200 where the retrieved context lacked the specific course information needed to answer the question. **Hallucination failures (n=3):** The model fabricated information not present in the retrieved context, and the fabricated information was factually incorrect. Examples include: hallucinating a quote from CS 111's description; providing prerequisite requirements for CS 221 that weren't in the context; and claiming prerequisite information was present when it wasn't. Notably, these cases had low answer_correctness scores (0.0-0.5), indicating the fabricated information was incorrect. This 5.7% hallucination rate (3 of 53 cases) includes all evaluated responses, both substantive answers and appropriate refusals, in the denominator. **Incomplete answers (n=1):** Responses omitted key information or provided partial answers when complete information was available. Example: when asked "When is CS 304 offered?", the system provided year-level information (third year) instead of the specific term offerings (Winter/Summer) that were requested, even though the context did not contain term scheduling information. **Over-generalization (n=1):** Responses provided generic guidance rather than specific policy interpretation. Example: when asked to "Tell me about CS", the system provided an extensive generic overview of Computer Science programs instead of asking for clarification about what specific aspect was needed. While hallucination failures occurred, they were relatively rare (3 cases) compared to retrieval issues. Most failures involved the model correctly using information from its training data that wasn't in the retrieved context (faithfulness issues) rather than fabricating incorrect information. This suggests the RAG architecture provides some constraint, though not complete protection against hallucination.

5 Discussion

Our evaluation provides initial characterization of how a RAG-based chatbot behaves across different types of academic advising queries. We discuss the implications for understanding system behavior, metric properties, and practical deployment, while acknowledging the limitations of automated evaluation.

5.1 How Reliable is AI Academic Advising?

Hallucination failures were rare but not absent: 3 of 53 test cases (5.7%) involved the model fabricating information not present in the retrieved context. In academic advising, where incorrect information about prerequisites or requirements could delay graduation or increase costs, even a low hallucination rate carries meaningful risk. The RAG architecture, which instructs the model to answer strictly from retrieved context, appears to provide substantial constraint but does not eliminate fabrication entirely. However, reliability is not solely about avoiding hallucinations. The failure mode analysis reveals that retrieval misalignment was actually the most common issue (6 cases), followed by hallucinations (3 cases). A student receiving an incomplete answer about credit exclusions could make suboptimal course decisions even without the system hallucinating. These findings indicate that while RAG reduces hallucination risk, retrieval quality remains the primary reliability challenge.

5.2 Metric Behavior Across Complexity

Our results reveal that RAGAS-style metrics behave differently across query complexity categories. Answer relevancy remains a weak discriminator as the system produces relevant responses nearly always (0.957-0.989). This ceiling effect limits its utility for identifying improvement areas. Context relevancy proves most informative for diagnosing retrieval quality. The 0.096 range across categories suggests retrieval performance varies meaningfully with

query type. Practitioners should prioritize this metric when optimizing RAG pipelines. The inverse relationship between faithfulness and context relevancy for edge cases merits attention. When retrieved context is irrelevant (0.643), the system can still achieve high faithfulness (0.843) by appropriately declining to answer. This demonstrates that faithfulness captures response grounding rather than retrieval quality, which is an important distinction for evaluation design. However, this also means that a system which frequently refuses to answer could achieve artificially high faithfulness scores while providing limited practical value. Hence, practitioners should interpret faithfulness alongside answer correctness and relevancy. Answer correctness, which compares against expert references, shows consistent discrimination across categories (0.739-0.786). This reference-based metric may be most valuable for assessing end-to-end system quality when ground truth is available.

5.3 LLM-as-a-Judge Validity

Several observations inform the validity of LLM-as-a-judge for educational RAG evaluation. The judge successfully identified appropriate refusal behavior on edge cases, awarding high relevancy and faithfulness scores for responses that politely declined out-of-scope queries. This suggests the rubric design adequately captures domain-appropriate behavior beyond simple factual matching. The detailed notes provided by the judge (required in our prompt design) offer interpretable explanations for scores. For example, one edge case received the note: "The assistant correctly identified the out-of-scope question and refused to answer, though the retrieved context was entirely irrelevant to the query." Such transparency enables manual review and rubric refinement. However, there are potential limitations. Using Gemini for both generation (2.5 Flash) and evaluation (3 Flash) may introduce family-specific or self-enhancement biases. Prior work on LLM-as-a-judge has observed possible self-enhancement effects and other biases (e.g., position, verbosity), though these are generally smaller than the overall agreement between strong judges like GPT-4 and human raters [15]. Future work should validate scores against human academic advisor assessments.

5.4 Implications for Practice

Our findings suggest that practitioners building educational RAG systems should prioritize retrieval quality above generation tuning. Context relevancy showed the most variation across query categories, indicating that the retrieval component represents the primary bottleneck. Embedding model selection and chunk design likely offer greater returns than prompt engineering for response generation. The strong performance on edge cases also highlights the value of designing systems for appropriate refusal. Our chatbot maintained high faithfulness scores precisely because it declined to answer when retrieved context was irrelevant. Clear scope definition in system prompts enables this behavior and prevents the system from overreaching into areas where it lacks grounding. Finally, our results underscore the importance of structured evaluation. Aggregate metrics obscured meaningful category-level variation that only became apparent when we examined common, specific, and edge cases separately. Practitioners should invest in expert-validated reference answers and organize test cases by expected complexity to surface actionable insights about system weaknesses.

5.5 Limitations

Several limitations scope our findings: **Sample size:** With 55 questions (53 successful), statistical power for fine-grained analysis is limited. Effect sizes and patterns should be interpreted cautiously. **Single system:** We evaluated one RAG configuration at one institution. Results may not generalize to other architectures or academic contexts. **No human validation:** We did not validate LLM-as-a-judge scores against human academic advisor assessments. The reported metrics reflect automated evaluation only. **Judge model selection:** Using Gemini Flash for both generation and evaluation may introduce systematic biases not present with other judge models. **Temporal snapshot:** Evaluation occurred at a single point in time. System performance may vary with course catalog updates or model changes.

6 Conclusion

We evaluated a production RAG-based course advising chatbot to provide initial empirical characterization of system behavior across query types. Using AdvisingBench, a 55-question benchmark across three complexity categories, and LLM-as-a-judge methodology, we observed that:

(1) Hallucination failures were rare (3 of 53 cases, 5.7%), suggesting RAG architectures may meaningfully constrain models to retrieved content, though they do not eliminate fabrication risk entirely.
(2) RAGAS-style metrics vary meaningfully across query complexity, with context relevancy most discriminative and answer relevancy showing ceiling effects.
(3) Edge cases achieve high faithfulness through appropriate refusal despite low retrieval relevancy, demonstrating the system knows when not to answer.
(4) Retrieval misalignment was the most common failure mode (6 cases), indicating that retrieval quality remains the primary reliability challenge.

These findings suggest that RAG-based chatbots substantially constrain hallucination in academic advising contexts, though the 5.7% hallucination rate and 11.3% retrieval failure rate indicate that full reliability remains elusive. Practitioners should focus on retrieval quality and use structured evaluation across query categories to identify weaknesses. Future work should: validate LLM-as-a-judge scores against human academic advisor assessments to establish metric reliability in this domain; expand the benchmark to additional institutions and programs; investigate failure mode taxonomy across query categories; and develop educational-domain-specific evaluation metrics capturing advisory appropriateness beyond factual correctness. AdvisingBench and evaluation code will be made available upon publication.

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